INTRODUCTION TO INFORMATION GEOMETRY

Fisher Length and Replicator Speed

Week 7 · Lecture 2

Christopher Lee

Length of a curve of distributions

For a differentiable path $P(t) = (p_i(t))$, its Fisher speed is

$$\|P'(t)\|_F = \sqrt{\sum_i \frac{(p_i'(t))^2}{p_i(t)}}.$$

Integrating gives its length:

$$L_F(P) = \int_a^b \|P'(t)\|_F \, dt.$$

The Bernoulli line element

For $P_p = (p, 1-p)$,

$$ds^2 = \frac{dp^2}{p(1-p)}.$$

The distance between $p = a$ and $p = b$ is

$$d_F(a, b) = \left| \int_a^b \frac{dp}{\sqrt{p(1-p)}} \right|.$$

The metric's varying coefficient controls the accumulated length.

An arc coordinate makes length constant

Set $p = \sin^2(\theta/2)$. Then

$$dp = \frac{1}{2}\sin\theta\, d\theta, \qquad \sqrt{p(1-p)} = \frac{1}{2}\sin\theta.$$

Thus $ds^2 = d\theta^2$, and

$$\theta(p) = 2\arcsin\sqrt{p}.$$

A Bernoulli Fisher distance

Use

$$d_F(a, b) = 2 \left| \arcsin \sqrt{b} - \arcsin \sqrt{a} \right| .$$

Find the distance from $a = \frac{1}{4}$ to $b = \frac{3}{4}$.

A Bernoulli Fisher distance

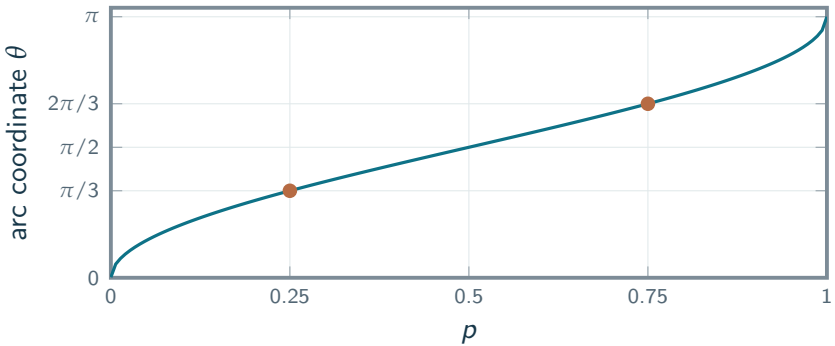
Use

$$d_F(a,b) = 2\left|\arcsin\sqrt{b} - \arcsin\sqrt{a}\right|.$$

The arc coordinates are $\theta_a = \frac{\pi}{3}$ and $\theta_b = \frac{2\pi}{3}$. Therefore

$$d_F(\tfrac{1}{4},\tfrac{3}{4}) = \frac{\pi}{3}.$$

The arc coordinate is nonlinear



The categorical Fisher matrix

Let $P = (x, y, z)$ with $z = 1 - x - y$. Then $dz = -dx - dy$, so

$$ds^2 = \frac{dx^2}{x} + \frac{dy^2}{y} + \frac{(dx + dy)^2}{z}.$$

In independent coordinates (x, y),

$$G(x, y) = \begin{pmatrix} 1/x + 1/z & 1/z \\ 1/z & 1/y + 1/z \end{pmatrix}.$$

The metric at a categorical law

For $P = (x, y, z) = (.2, .3, .5)$, compute

$$G = \begin{pmatrix} 1/x + 1/z & 1/z \\ 1/z & 1/y + 1/z \end{pmatrix}.$$

What are its three distinct entries?

The metric at a categorical law

For $P = (x, y, z) = (.2, .3, .5)$, compute

$$G = \begin{pmatrix} 1/x + 1/z & 1/z \\ 1/z & 1/y + 1/z \end{pmatrix}.$$

The result is

$$G = \begin{pmatrix} 7 & 2 \\ 2 & 16/3 \end{pmatrix}.$$

Tangent velocity as a random variable

At positive P, assign a tangent vector v the values

$$a_i = \frac{v_i}{p_i}.$$

Then $E_P[a] = \sum_i v_i = 0$, and

$$g_P(v, v) = E_P[a^2] = \mathrm{Var}_P(a).$$

Fisher norm is an ordinary variance when expressed in relative rates.

Replicator velocities have a simple norm

The replicator equation is

$$p_i' = p_i(f_i - \bar{f}).$$

Its relative rate is $p_i'/p_i = f_i - \bar{f}$. Hence

$$\|\mathbf{p}'\|_F^2 = \sum_i p_i(f_i - \bar{f})^2 = \mathrm{Var}_{\mathbf{p}}(f).$$

The speed of a replicator state

At $P = (.2, .3, .5)$, let $f = (3, 1, 2)$. The mean fitness is $\bar{f} = 1.9$. Compute

$\|P'\|_F^2 = \mathrm{Var}_P(f)$.

The speed of a replicator state

At $P = (.2, .3, .5)$, let $f = (3, 1, 2)$. The fitness deviations are $(1.1, -.9, .1)$.

$$\|P'\|_F^2 = .2(1.1)^2 + .3(-.9)^2 + .5(.1)^2 = .49.$$

The Fisher speed is $.7$.

Motion and geometry now use the same units

The replicator equation supplies tangent vectors. Fisher information assigns lengths to those vectors. The next step is to find the shortest intrinsic paths between two distributions. Their geometry becomes visible after a square-root change of coordinates.